WORKED SOLUTION

STEP III Specimen paper, Question 15

Probability and Statistics • Random Network Connectivity

STEP III Specimen paper, Question 15

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s3-q15

The key idea

Forget the geometry: all that matters is which of the six roads are open. Every village is reachable from every other exactly when the open roads form a *connected* network on the four villages. So we sort the possible sets of open roads by how many roads they contain, count how many of each size are connected, and weight each by its probability. With only four villages, the counting is short and clean.

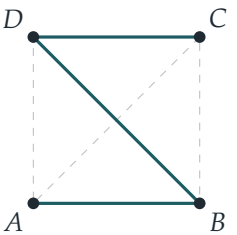
Solution

Write $q = 1 - p$ for the probability that a given road is open. Let K be the event that every village can be reached from every other after the snowfall.

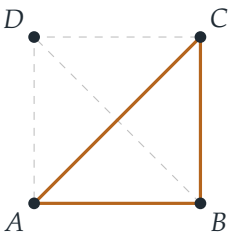
Since the six roads behave independently, any particular set of k open roads (with the other $6 - k$ blocked) has probability $q^k p^{6-k}$. So

$$\mathrm{P}(K) = \sum_{k=0}^{6} N_k \, q^k p^{6-k},$$

where N_k is the number of sets of k roads that connect all four villages.



three roads: a connecting tree



three roads: a triangle, D cut off

Counting the connected sets of roads

At most two open roads. Each open road joins two villages, so starting from one village, each road can bring in at most one new village. Reaching all three others needs at least three roads. So $N_0 = N_1 = N_2 = 0$.

Three open roads. There are $\binom{6}{3} = 20$ sets of three roads. Suppose such a set fails to connect the villages. If some village is cut off, all three roads lie among the other three villages. But only three roads join those three villages, so the set is exactly a triangle. If no village is cut off, the villages split into two pairs, and at most two roads (one inside each pair) can be open, a contradiction. So the only failures are the 4 triangles, one for each choice of the cut-off village, and

$$N_3 = 20 - 4 = 16.$$

? **Ask yourself** Why can't four open roads ever leave the network disconnected?

Four or more open roads. Use the same two ways of failing. A cut-off village would force all the open roads to lie among the other three villages, where there are only three roads. A split into two pairs allows at most two open roads. With four or more open roads, neither can happen, so every such set is connected:

$$N_4 = \binom{6}{4} = 15, \qquad N_5 = \binom{6}{5} = 6, \qquad N_6 = 1.$$

The probability

Putting the counts together,

$$\mathrm{P}(K) = q^6 + 6q^5p + 15q^4p^2 + 16q^3p^3, \qquad q = 1 - p.$$

Expanding in powers of p gives the equivalent form

$$\mathrm{P}(K) = 1 - 4p^3 - 3p^4 + 12p^5 - 6p^6.$$

The case $p = \frac{1}{2}$

Now $p = q = \frac{1}{2}$, so every one of the $2^6 = 64$ patterns of open and blocked roads is equally likely, with probability $\frac{1}{64}$. Hence

$$\mathrm{P}(K) = \frac{1 + 6 + 15 + 16}{64} = \frac{38}{64} = \frac{19}{32},$$

as required.

Checks

- *Extreme values.* At $p = 0$ every road is open and the formula gives 1. At $p = 1$ every road is blocked and it gives $1 - 4 - 3 + 12 - 6 = 0$.

- *The complement.* The network is disconnected exactly when some village is cut off, or the villages split into two pairs with no road between the pairs. A village is cut off with probability p^3. By inclusion–exclusion over the four villages (two particular villages cut off needs five blocked roads, and three cut off means all four are) this gives $4p^3 - 6p^5 + 3p^6$. Each of the 3 splits into pairs, with both inside roads open, adds p^4q^2. One minus the total agrees with the formula above: at $p = \frac{1}{2}$ the disconnected probability is $\frac{32-12+3+3}{64} = \frac{13}{32}$.
- *Brute force.* Listing all 64 sets of roads by computer confirms the counts $16, 15, 6, 1$. A simulation with $p = 0.3$ gave 0.893, against the exact value 0.8925.

Common slips

- Counting all 20 sets of three roads as connected, forgetting the four triangles.
- Assuming any set of four roads can still leave a village isolated. A village is cut off only if all three of its roads are blocked.
- Mixing up p (blocked) and $1 - p$ (open) when writing $q^k p^{6-k}$.

Beyond STEP

This is a tiny case of a *random graph*: each edge of a network survives independently, and we ask whether the result is connected. The polynomial in p is called the *reliability polynomial* of the network, and the number 16 is an instance of Cayley's formula: there are n^{n-2} trees joining n labelled points. For large networks, Erdős and Rényi found that connectivity switches on remarkably suddenly as the open-road probability passes a threshold. For more on random structures built from independent events, see Grimmett and Welsh, *Probability: An Introduction*.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.